

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

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ABSTRACT

The scarcity of annotated data in specialized domains such as medical imaging is a fundamental obstacle to training robust vision models. Self-supervised masked image modeling (MIM) is the natural remedy, yet existing approaches remain shackled to random high-ratio masking, which is both inefficient and semantically misaligned. Region-aware variants do not solve the problem: they depend on reconstruction heuristics or supervised signals that cripple their adaptability across tasks and modalities. We introduce Mask What Matters, a controllable text-guided masking framework for self-supervised medical image analysis that removes these constraints. By harnessing vision-language models for prompt-based region localization, our method applies differentiated masking that sharply emphasizes diagnostically relevant regions while stripping away redundancy in background areas. This controllable design delivers markedly better semantic alignment, stronger representation learning, and robust cross-task generalizability. Comprehensive evaluation across three medical imaging modalities—brain MRI, chest CT, and lung X-ray—establishes that Mask What Matters consistently and decisively outperforms existing MIM methods (e.g., SparK), achieving gains of up to +3.1 percentage points in classification accuracy, +1.3 in box average precision (BoxAP), and +1.1 in mask average precision (MaskAP) for detection. Crucially, it attains these gains while masking far less of the image (e.g., 40% vs. 70%). Taken together, these results establish that controllable, text-driven masking is an effective route to semantically aligned self-supervised learning, and chart a clear path toward robust vision models for medical image analysis.

1 INTRODUCTION

Image classification, segmentation, and object detection are the fundamental tasks of computer vision, and deep learning is now the core technology driving progress across all of them. Yet high-performing deep learning models depend on large-scale annotated datasets. In specialized domains such as medical image processing, high-quality annotations are scarce and expensive, and this single bottleneck stands as the decisive barrier to the widespread adoption of deep learning methods.

Self-supervised learning (SSL) directly addresses this bottleneck: by pretraining on unlabeled data, it lets models learn effective feature representations without any manual annotation [Jaiswal et al. (2020); Liu et al. (2021); Krishnan et al. (2022); Huang et al. (2023)]. Among SSL paradigms, MIM has become the dominant approach. By masking parts of an image and forcing the model to reconstruct the original content, MIM compels the model to capture structural and semantic patterns from visible context. It delivers strong performance across a wide range of downstream tasks and transfers readily to medical image analysis [He et al. (2022); Gupta et al. (2025); Xiao et al. (2023)].

However, prevailing MIM approaches—such as masked autoencoders (MAE) [He et al. (2022)] and SparK [Tian et al. (2023)]—rely on random masking strategies that are fundamentally blind to the regions of interest in downstream tasks. This task-agnostic masking imposes two decisive limitations: (1) The learned representations are semantically misaligned with the downstream objectives, which directly erodes transfer performance. (2) To have any chance of masking informative areas, these methods resort to extremely high masking ratios (e.g., 75% in MAE), inflating reconstruction

difficulty and demanding large-scale data and computation. Natural language stands in sharp contrast: its compact structure and high semantic density enable effective learning even at low masking ratios [Devlin et al. \(2019\)](#). The dependence on high masking ratios in image-based MIM is thus a direct symptom of information redundancy and the absence of semantic-aware masking, which together prevent models from focusing on the regions that truly matter.

This issue is especially acute in medical imaging. Medical images exhibit extreme semantic sparsity: diagnostically relevant information is confined to small localized regions (e.g., lesions or organs), while the overwhelming majority of the image is semantically redundant background. Medical imaging further spans diverse modalities—MRI, CT, and X-ray—with substantial variation in the shape, size, and spatial distribution of task-relevant areas across tasks. Together, these properties make efficient, generalizable masking not merely desirable but essential.

We resolve these challenges with a controllable, text-guided masking framework—**Mask-What-Matters (MWM)**. Given a user-specified, open-vocabulary description or phrase, a pretrained vision-language model supplies zero-shot localization cues that we convert into robust regions of interest (ROIs). MWM then applies region-specific, prompt-conditioned masking—assigning higher ratios to semantically important areas (e.g., lesions/organs) and lower ratios to background—thereby injecting downstream semantics directly into the pretraining signal, and doing so without any per-image reports or labels.

To summarize, our main contributions are:

- **The first controllable text-guided masking framework for medical imaging.** MWM integrates vision-language models (BiomedCLIP) with segmentation refinement (SAM) to localize task-relevant regions from open-vocabulary prompts and apply differentiated ROI vs. background masking, directly eliminating the semantic misalignment that plagues random masking.
- **An annotation-free and backbone-agnostic design.** MWM requires no per-image reports or labels, and integrates seamlessly into both ViT- and ConvNet-based MIM pipelines under a single unified protocol.
- **Consistent gains across modalities and tasks.** On brain MRI, chest CT, and X-ray datasets, MWM decisively surpasses state-of-the-art MIM baselines (e.g., SparK), improving classification, detection, and segmentation alike, and establishing text-driven generative pretraining as a powerful paradigm for medical imaging.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling originated in natural language processing with the bidirectional encoder representations from transformers (BERT) model [Devlin et al. \(2019\)](#), which learns contextual representations by masking tokens and predicting them during pretraining. Building on this idea, [He et al. \(2022\)](#) brought MIM to computer vision, masking portions of an image during pretraining and tasking the model with reconstructing the missing regions. This paradigm generalizes strongly across both natural and medical image domains [\(Gupta et al., 2025\)](#) [\(Liu et al., 2021\)](#) [\(Huang et al., 2023\)](#). Representative methods include MAE [\(He et al., 2022\)](#), MedMAE [\(Gupta et al., 2025\)](#) [\(Xiao et al., 2023\)](#), and SparK [\(Tian et al., 2023\)](#). MAE uses high-ratio random masking to push ViTs toward high-level semantic representations; MedMAE extends this to medical images with domain-specific adaptations; and SparK adapts the strategy to convolutional backbones, exploiting their multi-scale features to strengthen pretraining.

Despite these advances, conventional MIM approaches rely on random or structure-agnostic masking that ignores the semantic requirements of downstream tasks. Such non-selective masking forces the model to squander capacity on irrelevant regions during pretraining, directly reducing transfer effectiveness. Recent work has therefore turned to region-aware masking to restore semantic alignment, along two main directions:

One line of work introduces loss-driven adaptive masking: methods such as hard patches mining (HPM) [\(Wang et al., 2023a\)](#) and its extension AnatoMask [\(Li et al., 2024\)](#) analyze reconstruction errors across regions and assign higher masking probabilities to high-loss areas, aiming to prioritize semantically valuable structures. This strategy has a fundamental flaw—it conflates hard-to-

reconstruct with important, and thus misdirects the model toward difficult but task-irrelevant regions. In brain magnetic resonance imaging (MRI), for instance, complex structures such as the spinal cord or eyeballs draw attention away from the downstream target of tumor detection.

A second line of work adds external perceptual modules to guide masking. FocusMAE [Basu et al. 2024], for example, uses pretrained Region Proposal Networks (RPNs) to identify high-information regions for selective masking. Effective as it is on specific tasks, this strategy requires supervised training of RPNs on domain-specific annotated data—precisely the annotation dependency that undermines cross-task generalization.

These limitations point unmistakably to the need for a masking mechanism that is at once flexible, semantically aligned, and generalizable. Natural language, a rich and interpretable source of supervision, is exactly such a signal: it can guide masked modeling in a task-aware yet fully annotation-free manner.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

The rapid maturation of vision-language alignment models has made it possible to accurately perceive and identify task-relevant regions under zero-shot and open-world settings [Radford et al. 2021]. The Contrastive Language-Image Pretraining (CLIP) family builds a unified embedding space for images and texts through large-scale joint pretraining on image-text pairs, and is now standard for zero-shot and open-set visual tasks. In the medical domain, extensions such as BiomedCLIP [Zhang et al. 2023] sharpen cross-modal understanding and deliver robust performance across diverse downstream applications. Yet despite this success, the use of vision-language models has been confined almost entirely to supervised downstream tasks such as object detection and semantic segmentation [Zhong et al. 2022, Aleem et al. 2024, Koleilat et al. 2024]. Their open-vocabulary, zero-shot capabilities remain untapped for masked image modeling—a gap our work is the first to close.

In summary, recent region-aware MIM approaches—dynamic masking based on reconstruction loss, or masking guided by external perception modules—improve semantic focus only partially, and each remains hobbled by at least one of three shortcomings: dependence on additional supervision, misalignment between masked regions and task semantics, and a lack of flexibility. At the same time, the strong open-vocabulary abilities of vision-language models have simply not been brought to bear on the MIM pretraining phase. This gap defines a clear and compelling opportunity: to inject vision-language semantic guidance into masking and thereby make masked image modeling both efficient and generalizable—the opportunity our work seizes.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

As illustrated in Figure 1, **Mask What Matters (MWM)** is built from two core stages: (1) text-guided region localization and (2) region-aware masked image modeling. The first stage uses a frozen vision-language model to pinpoint task-relevant regions from an open-vocabulary prompt; the second applies differentiated masking ratios across those regions to steer the reconstruction process.

3.2 TEXT-GUIDED REGION LOCALIZATION

MWM localizes task-relevant regions through a purpose-built multi-stage pipeline. It begins with LLM-based prompt generation and uses BiomedCLIP to extract cross-modal embeddings. A saliency map is then produced via the Multi-Modal Information Bottleneck (M2IB), binarized with K-Means, and refined with the Segment Anything Model (SAM). Finally, the refined mask is converted into a bounding box with a controllable expansion margin, yielding a robust spatial guide for the masked modeling stage that follows.

Prompt Generation. In medical image analysis, prompt design is decisive for the quality of region localization. To maximize semantic guidance, we query large language models (LLMs) with structured prompts such as “Describe the typical visual characteristics of a [category] in a [modality]

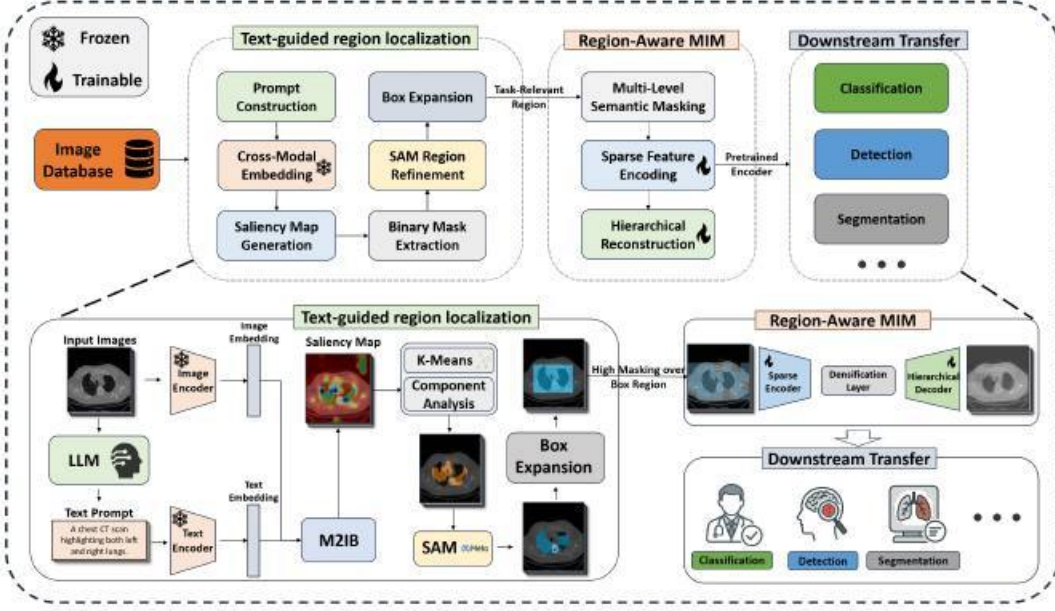


Figure 1: Overview of the MWM framework. The top panel lays out the three-stage pipeline: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The bottom panel zooms in to show exactly how prompts drive region localization and masking.

image.” We also evaluate simple category phrases as an alternative input, and present a detailed comparison of the two prompt styles in the experimental section.

Cross-Modal Embedding. Once the prompt is generated, a BiomedCLIP model [Zhang et al. \(2023\)](#) encodes the medical image and its prompt into a shared semantic space. Given an image I and a prompt T , the embeddings are computed as:

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. These cross-modal embeddings form the semantic foundation on which the subsequent saliency estimation is built.

Saliency Map Generation. We employ a cross-modal saliency estimation module based on the Multi-modal Information Bottleneck (M2IB) [Wang et al. \(2023b\)](#), which localizes task-relevant visual regions by aligning image and text embeddings while suppressing modality-specific redundancy. Formally, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ assigns each spatial location an importance weight reflecting its semantic relevance to the prompt. M2IB obtains this map by optimizing the following objective:

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} are image and text embeddings encoded by BiomedCLIP respectively, and γ controls the trade-off between preserving cross-modal relevance and filtering out task-irrelevant visual information.

Binary Mask Extraction. Given the saliency map, we binarize it with unsupervised K-Means clustering [Lloyd \(1982\)](#) to localize the text-guided region and produce a preliminary foreground mask. The map is clustered into two pixel groups, and the cluster with higher saliency is taken as foreground. To suppress false activations, we then apply connected component analysis to the binary map and retain only the largest foreground regions. This step reliably filters out noisy edge areas and concentrates attention on the core semantic structures.

Region Refinement. To sharpen spatial accuracy, we incorporate SAM as a refinement module [Kirillov et al. \(2023\)](#). For each selected connected region $c_i \in \mathcal{C}^*$, we compute its minimal enclosing bounding box and use it as the visual prompt for SAM.

Given the original image I and the set of bounding boxes, SAM predicts a refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

The resulting mask M_{SAM} captures the spatial structure far more faithfully.

Box-Based Region Expansion. The SAM mask still carries boundary noise that can mislead the model’s focus during representation learning. To make region guidance robust, we convert the refined SAM mask into a bounding box with a controllable expansion margin, which serves as the final region of interest (ROI) for downstream masked modeling.

3.3 REGION-AWARE MASKED IMAGE MODELING

Given the region of interest produced by the localization module in Section 3.2, MWM performs masked image modeling through a three-stage pipeline: (1) a multi-level masking strategy that applies differentiated masking ratios according to region importance, (2) a sparse encoder that processes only the unmasked patches, and (3) a hierarchical decoder that reconstructs the full image from sparse multi-scale features.

Multi-Level Semantic Masking. MWM adopts a multi-level masking strategy that directly exploits the semantic regions identified by the text-driven localization process. Patches within regions highlighted by the textual prompts—those associated with tumors or organs—are masked at a high ratio, forcing the model to concentrate on reconstructing task-relevant semantic features. Background regions, by contrast, are masked lightly or left unmasked to cut redundant computation. Figure 2 contrasts the resulting masking patterns.

Sparse Feature Encoding. Visible patches are collected into a sparse image and encoded following the SparK design [Tian et al. \(2023\)](#):

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where x_i is the i -th patch, $m_i \in \{0, 1\}$ indicates visibility, Φ_{sparse} is the encoder, and $F = \{F_1, F_2, F_3, F_4\}$ are multi-scale features. We instantiate MWM with convolutional backbones, but the same interface applies directly to ViTs via token indexing.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the full image top-down. At each scale l , masked locations are filled with a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified map F'_l is projected by a scale-specific layer ϕ_l to obtain D_l , then upsampled and fused with features at the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

Finally, the model is trained with mean-squared error on masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

where $\mathcal{I}_{\text{mask}}$ is the set of masked patch indices, x_i the original patch, and $\hat{x}_i$ its reconstruction.

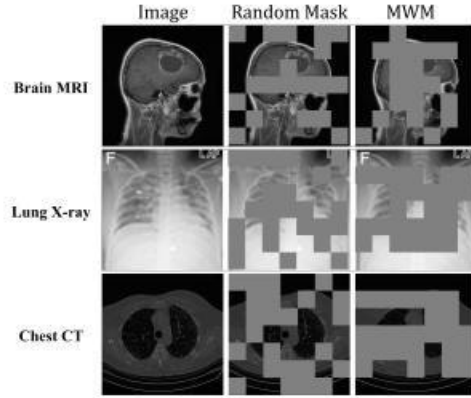


Figure 2: Comparison of masking strategies. Gray blocks indicate masked regions.

4 EXPERIMENTS

4.1 DATASETS

Self-Supervised Pretraining Datasets. We pretrain on three datasets spanning distinct imaging modalities. (1) *Brain MRI* includes approximately 17,000 images from Brain Tumor MRI [Cheng et al. (2015)], BRISC [Fateh et al. (2025)], and BraTS 2018 [Menze et al. (2014)], covering various tumor types and anatomical structures. (2) *Lung X-ray* consists of around 39,000 chest radiographs from COVID-QU-Ex [Rahman et al. (2021)]; Chowdhury et al. (2020), including COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT* comprises 12,000 slices from a lung disease dataset [Konya (2020)], with segmentation masks of fibrotic lungs from 107 patients.

Downstream Fine-Tuning Datasets. We evaluate on three downstream datasets spanning classification and detection tasks. (1) *Brain Tumor MRI* [Feltrin (2023)] includes 4,479 MRI images spanning 44 tumor types, such as astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* [Kermany et al. (2018)] consists of 5,863 pediatric chest X-rays categorized as either normal or pneumonia. (3) *Chest CT Cancer* [Hany (2020)] includes CT images from four classes: adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue.

4.2 EXPERIMENTAL SETUP

To guarantee a fair comparison, all self-supervised methods use backbones of comparable scale: convolution-based methods adopt ResNet-50 [He et al. (2016)] and transformer-based methods use ViT-S [Dosovitskiy et al. (2020)]. The input resolution is 224×224, and models are initialized with ImageNet-pretrained weights. Pretraining uses an initial learning rate of 2×10^{-4} , and downstream fine-tuning retains the same encoder and input resolution. All experiments run on NVIDIA Tesla A100 cards with 80 GB VRAM under Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless otherwise specified, we fix non-target variables (e.g., prompt type or SAM usage) to their empirically optimal settings, ensuring every comparison is strictly controlled.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first establish the effectiveness of the text-guided localization module across three imaging modalities—brain MRI, lung X-ray, and chest CT—and then dissect the impact of prompt type and SAM refinement. We compare predicted bounding boxes against expert-annotated segmentation masks, reporting occlusion precision ($|R_p \cap R_{gt}| / |R_p|$) and occlusion recall ($|R_p \cap R_{gt}| / |R_{gt}|$), where R_p is the predicted region and R_{gt} is the ground-truth annotation.

As Table 1 shows, with effective prompts our text-guided localization attains consistently high occlusion recall across all datasets (around 0.82) while holding precision at a solid level. This demonstrates that the framework identifies task-relevant regions from textual descriptions alone, and does

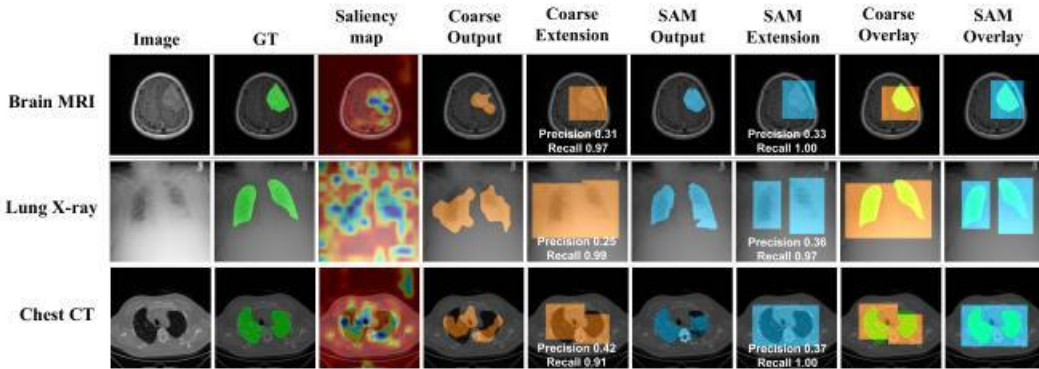


Figure 3: Representative results of text-guided region localization, showing accurate localization across modalities.

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Localization performance (Occlusion Precision/Recall) across prompt types and refinement strategies. *Coarse Ext.* denotes bounding box extension of the coarse region; + *SAM Ext.* adds further refinement with SAM, which consistently sharpens precision.

so robustly despite pronounced modality-specific anatomical variation (see Figure 3 for visualized examples).

Effect of Prompt Type on Region Localization. We next isolate the role of language design by comparing two prompt types. As Table 1 shows, descriptive sentence prompts win on Brain MRI and Chest CT, confirming that richer semantics sharpen localization. Concise phrases, in turn, take the lead on lung radiographs in recall (0.82 vs. 0.66)—a direct consequence of the lungs’ broad anatomical coverage, for which general terms are already sufficient.

Effect of SAM Refinement on Region Localizatio. We then quantify the contribution of spatial refinement by applying SAM to the coarse masks. As Table 1 shows, SAM refinement raises precision while preserving high recall. On Brain MRI, precision jumps from 0.22 to 0.30 with recall held above 0.94. SAM thus clearly refines coarse localization outputs at no cost to coverage.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

We now benchmark the generalization and representation ability of MWM against representative self-supervised learning (SSL) methods spanning both reconstruction-based and contrastive paradigms. Every method is pretrained and evaluated under identical datasets and training settings, so the comparison is unambiguously fair.

Image Classification. We examine classification performance on three downstream medical imaging datasets. Following common practice, we report two settings: (1) *full fine-tuning*, where the entire model is updated; and (2) *linear probing*, where the encoder is frozen and only a linear classifier is trained on top.

- **Full fine-tuning:** Table 2 summarizes classification accuracy for MWM and several representative self-supervised methods, including MAE [He et al. (2022)], SparK [Tian et al. (2023)], AnatoMask [Li et al. (2024)], SimCLR [Chen et al. (2020a)], MoCoV2 [Chen et al. (2020b)], and

Method	Type	Brain MRI	Chest CT	Lung X-ray
MoCoV2	CL	95.3	91.7	93.1
BYOL	CL	91.3	93.7	93.9
SimCLR	CL	94.5	93.1	94.1
MAE	MIM	95.4	91.7	94.9
AnatoMask	MIM	96.5	95.8	95.2
SparK	MIM	96.2	94.4	94.7
MWM(Ours)	MIM	96.8	97.5	96.0

Table 2: Fine-tuning classification accuracy (%) on downstream datasets, where MIM denotes Masked Image Modeling and CL denotes Contrastive Learning. MWM attains the best accuracy on every modality.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Linear probing accuracy (%) on downstream datasets (frozen encoder). MWM leads on all three modalities.

Method	Detection		Segmentation	
	AP ^{box}	AP ^{box} ₇₅	AP ^{mask}	AP ^{mask} ₇₅
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: Detection (AP^{box}, AP^{box}₇₅) and instance segmentation (AP^{mask}, AP^{mask}₇₅) results (%) on the BR35H dataset. MWM leads on detection and remains highly competitive on segmentation.

BYOL [Grill et al. \(2020\)](#). MWM achieves the highest accuracy on every modality, establishing strong and consistent generalization across diverse imaging domains. The gain is largest on Chest CT, where MWM reaches 97.5% accuracy—surpassing SparK by +3.1 and AnatoMask by +1.7. This result pinpoints the benefit of text-guided masking: it emphasizes semantically critical yet spatially sparse regions—such as lung lobes in CT scans—while eliminating masking redundancy.

- **Linear probing:** To probe representation quality directly, we run linear probing against two recent state-of-the-art masked image modeling methods—SparK and AnatoMask. As Table 3 shows, MWM outperforms both on all datasets by clear margins, confirming that text-driven pretraining yields substantially more effective features.

Object Detection and Instance Segmentation. To test whether the learned representations generalize beyond classification, we evaluate brain tumor detection and instance segmentation on the BR35H dataset [Hamada \(2025\)](#). As Table 4 shows, MWM achieves the highest AP^{box} (46.9%) and a strong second-best AP^{mask} (45.9%), surpassing SparK by +1.3 and +1.1 points, respectively. MWM therefore transfers effectively to both object-level localization and pixel-wise segmentation.

In short, injecting text-driven masking into pretraining yields markedly more informative and transferable representations. This semantic guidance improves performance across classification, detection, and segmentation alike, confirming its value for visual representation learning.

4.5 ABLATION STUDY

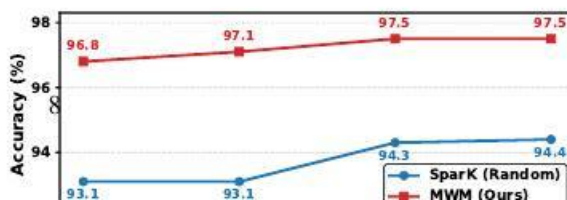
We further conduct ablation experiments that isolate the individual impact of prompt design, SAM-based region refinement, and masking ratio on MWM’s downstream classification performance.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

As Table 5 shows, descriptive sentence prompts outperform short phrases on brain MRI and chest CT, while the reverse holds on lung X-ray. This tracks the localization results in Section 4.3 exactly: prompts that localize regions more accurately also produce better downstream results. Crucially, both prompt types—regardless of granularity—beat random masking on every dataset. This establishes that text-guided masking delivers effective semantic alignment between pretraining and downstream tasks, the key driver of improved self-supervised performance. The SAM ablation reinforces the point: incorporating SAM during pretraining consistently lifts downstream classification, confirming that it strengthens text-guided masking.

4.5.2 EFFECT OF MASKING RATIO

As Figure 4 shows, MWM reaches 96.8% classification accuracy at a masking ra-



Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Downstream classification accuracy (%) under different prompt types and region refinement strategies across three imaging modalities. *No Prompt* refers to random masking without text-guided localization. *w/* and *w/o* denote with and without SAM, respectively. Both text guidance and SAM refinement consistently help.

ratio of just 40%—far below the 70% that masked image modeling typically demands. Even at this reduced ratio, it beats SparK’s best 70%-masking result by +2.4% in absolute accuracy. To rule out that SparK’s weaker showing is merely a matter of suboptimal masking, we evaluate SparK at 40%, 50%, 60%, and 70%. The 70% setting is indeed SparK’s best, which confirms that our comparison is fair and that MWM’s advantage is genuine. These results overturn the prevailing assumption that high masking ratios are inherently optimal for masked image modeling. They show instead that semantically guided, task-aware masking—as in MWM—achieves stronger downstream performance at lower masking ratios by preserving task-relevant content and stripping away redundancy.

5 CONCLUSION AND LIMITATIONS

This work introduces **Mask What Matters (MWM)**, a self-supervised pretraining framework that unifies text-guided semantic localization with region-aware masking. By using natural language prompts to highlight task-relevant areas and applying differentiated masking, MWM delivers semantically aligned representation learning without any per-image annotation.

Comprehensive experiments establish the effectiveness and generalizability of MWM: (1) natural language prompts reliably guide semantic region localization, proving text-driven masking is not only feasible but effective; (2) MWM consistently outperforms existing methods (e.g., SparK, AnatoMask) in classification across all three imaging modalities, and delivers further gains in detection and instance segmentation; (3) MWM sustains strong performance even at low masking ratios (e.g., 40%), demonstrating the decisive advantage of semantic guidance in self-supervised pretraining.

Limitations. Two limitations bound the present study and define concrete next steps. First, while text prompts provide flexible guidance, we have not yet systematically characterized MWM’s robustness to variations in prompt style, vocabulary, or noise. Second, the reliance on external textual descriptions is a dependency that may weaken generalization when reliable prompts are unavailable or inconsistent.

Looking forward, text-driven, region-aware masking offers a principled route to more semantically grounded self-supervised learning, and extending MWM beyond medical imaging is a natural and promising avenue for broader vision–language pretraining.

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A USE OF LARGE LANGUAGE MODELS (LLMs)

We used LLMs *only* for language editing and presentation polishing (e.g., grammar, phrasing, and minor stylistic clarity) of text written by the authors. LLMs were *not* used for idea generation, experimental design, data annotation, code implementation, or analysis/interpretation of results. All technical content, algorithms, proofs, figures, tables, metrics, and conclusions were authored and verified by the authors. No confidential or identifying data were provided to LLMs beyond anonymized manuscript text.